

Similar System Data Set for Fault Diagnosis: Bearing Faults with Different Bearing Types

Authors: Braig, Marcel; Neu, Moritz; Zeiler, Peter
Institution: Esslingen University of Applied Sciences,
Institute for Technical Reliability and Prognostics (IZP)
Contact: IZP@hs-esslingen.de

Date: June 12, 2025

Description of the test rigs:

This data set originates from two test rigs for bearing fault diagnosis. On the test rigs, pre-damaged bearings with different fault types are installed and rotated with a set rotation speed. Rotation speed and vibration acceleration are recorded over time, enabling the diagnosis of the faults of installed bearings. The test rigs are constructed as seen in Figure 1. The test rig on the left side is referred to as test rig (a), the other one as test rig (b). Test rig (b) has a special socket for the diagnosed bearing, which offers the possibility to install bearings of different types. The data set is uploaded to Kaggle by the user "Prognostics @ HSE".

Because the measurements were taken on two different test rigs and with different bearing types, the given data set includes data of so-called similar systems with similar but not identical technical characteristics. This makes it particularly interesting for use in the research field of transfer learning. It is therefore recommended to use the data set provided to investigate the transfer of information between different bearing types or the two test rigs. Due to the mechanics of a bearing can be modeled, it might also be beneficial to support the actual data-driven models by integrating physical knowledge, respectively, models in the sense of theory-guided data science or informed machine learning.

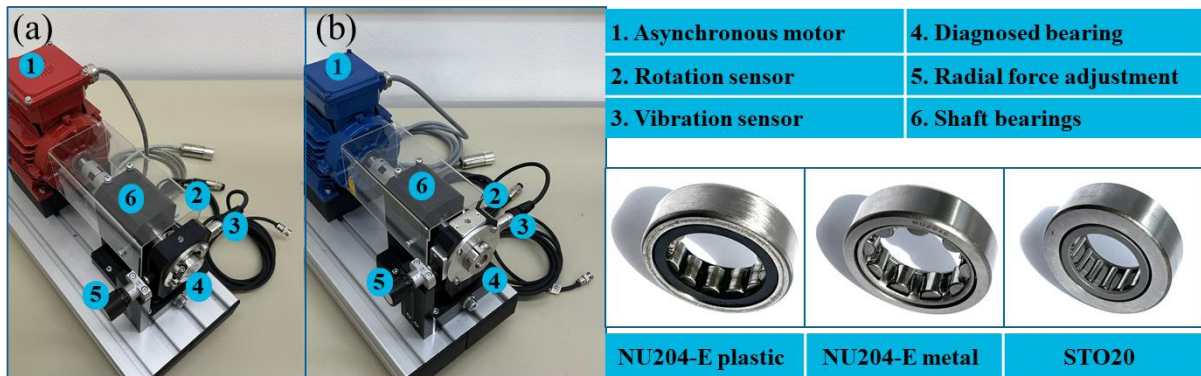


Figure 1: The test rigs and the different bearing types used

The asynchronous motor (1) is controlled to hold the set speed that can be monitored with the rotation sensor (2). The vibration sensor (3) is used to record the vibrations of the diagnosed bearing (4). The screw with the rubber bumper (5) allows the user to adjust the force that is applied to the diagnosed bearing. There is also a shaft with bearings (6) that connects the motor and the diagnosed bearing. The outer ring of the diagnosed bearing is fixed, whereas the inner ring rotates with the shaft.

As seen in the schematic illustration in Figure 2, the vibration sensor is positioned opposite the load zone. The vibrations of the diagnosed bearing, which arise primarily in the load zone, reach the sensor because it is connected with the outer ring of the bearing via the bearing socket.

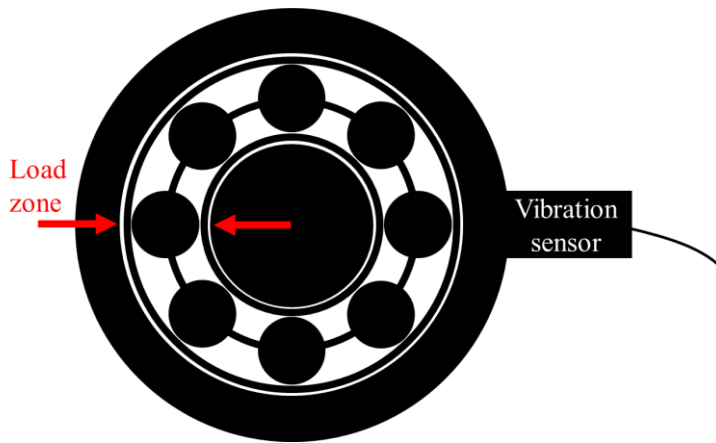


Figure 2: Front view on the diagnosed bearing on the test rig (schematic)

Diagnosed bearings:

The diagnosed bearing types that were used on two different test rigs to generate the data set are shown in Figure 1, and their parameters are given in Table 1. Note that the bearing type NU204-E with the plastic cage has been installed on both test rigs, which is why there are only three pictures shown. However, due to the different constructions of the bearing sockets on the test rigs (a) and (b), the two setups also differ from one another, resulting in different vibration signals, even if the exact same bearing is mounted. For the sake of simplicity, all four cases (three different bearing types and one bearing installed on two test rigs) are therefore simply referred to as four different “bearing types”.

Table 1: Bearing types with parameters and possible fault types

Model	Rig	d / mm	D / mm	D_{RE} / mm	# of RE	RE type	Cage	Fault types
NU204-E	a	20	47	7.5	12	Cylinder	Plastic	OK / IR / OR / RE
NU204-E	b	20	47	7.5	12	Cylinder	Plastic	OK / IR / OR / RE
NU204-E	b	20	47	7.5	9	Cylinder	Metal	OK / IR / RE
STO20	b	20	47	3.5	14	Needle	Metal	OK / IR / OR / RE

d : inner diameter, D : outer diameter, D_{RE} : diameter of rolling elements, # of RE: number of rolling elements

Measurement explanation:

Each of the four different bearing types in this data set can have up to four possible fault types. Possible fault types are: OK – healthy, IR – inner race fault, OR – outer race fault, RE – rolling element fault. As can be seen in Table 1, for one bearing type only three of the four fault types are present. The measurements were taken under two different speeds (500, 1000 1/min) and two different load levels. Five recordings with the length of 1 s each are taken for each combination of bearing type, fault type, speed, and load. This results in a total number of

$$(3 \text{ bearing types} \cdot 4 \text{ fault types} + 1 \text{ bearing type} \cdot 3 \text{ fault types}) \\ \cdot 2 \text{ speeds} \cdot 2 \text{ loads} \cdot 5 \text{ recordings} = 300$$

recordings.

Different bearing fault types:

The damage was done to the bearings by artificially removing material at one point on the surface of the corresponding components so that a furry was formed. So, for example, in the case of the fault type IR, a furry in the inner race was inserted. These furry with less material on the different components lead to impulses as the bearing rotates. According to the fault type, the bearing type, the speed, and the load, the impulses occur with different time intervals and amplitudes in the vibration measurements.

The characteristic fault frequencies can be calculated for each bearing type; the formulas can be found in the corresponding literature on rolling element bearing fault diagnosis.

Procedure for a measurement:

In the first step for recording a measurement, the bearing is mounted on the test rig, and the load level is adjusted by compressing the rubber bumper with the screw. Then the motor is turned on, and the speed is set with a potentiometer. As soon as the desired speed is set, the measurements are recorded. It should be noted that the angle of the fault when the recording of a measurement starts is random, which leads to a temporal shift in the vibration signals.

Given data situation:

The following settings were made for data acquisition:

- Length of recordings: 1 s
- Sampling frequency: 15 625 Hz
- Speed adjustment: by hand, with potentiometer

As mentioned above, the data set was recorded on two different test rigs, which have different sockets for the bearings. The vibration sensor is the same model on both test rigs. The control device was the same for both test rigs, and also the software that was used to perform the measurements was the same. Figure 3 shows two vibration signals over a cut-out of 0.1 s from two different diagnosed bearing types under the same rotation speed. As can be seen, both bearings have similar characteristics in the vibration signal, which makes information transfer by means of transfer learning promising.

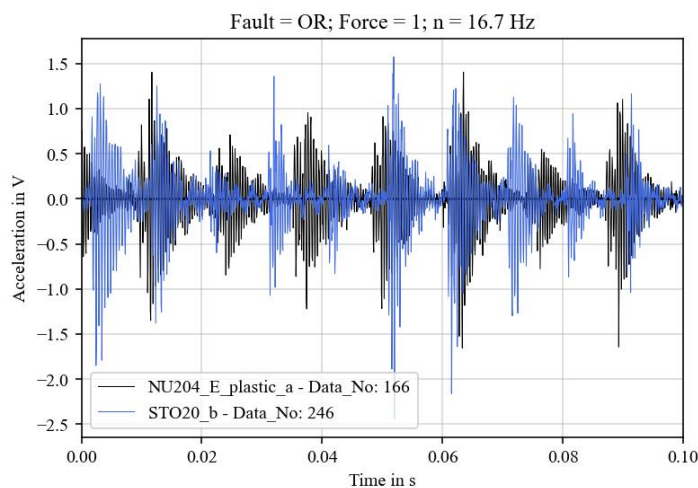


Figure 3: A 0.1 s cutout from two recordings of two different bearing types with the same fault type 'OR'

Task:

The goal is to determine the fault type of the diagnosed bearing by the given recordings. The main challenge regarding transfer learning is to utilize data from other bearing types in order to enhance the performance of the fault type classifier for the bearing under consideration. Due to the detailed description of the bearing parameters and the literature about rolling element bearing fault diagnosis, there is also the possibility of supporting the data-driven models by integrating physical knowledge.

Data structure:

The data set is stored in a .csv file, which is structured as follows. The first row contains a headline that describes which values are contained in the corresponding column. From the second row on, all entries in the .csv file are actual data. In Table 2, the meaning of each column headline is explained. Note that the value of 'Speed_Set' is given in 1/min, while the actual speed is given in Hz. This makes filtering for recordings at one of the two (set) speed levels easier.

Table 2: Description of the data in each column of the data set

Time	Time value of each measurement / s
Acceleration	Measured voltage from the vibration sensor / V
Fault	Fault type of the bearing (label) ['IR', 'OK', 'OR', 'RE']
Speed_Real	Speed, calculated from the signal of the rotation sensor / Hz
Speed_Set	Speed, that was set, which can be either 500 or 1000 / 1/min
Force	Load level, which can be either 1 or 2
Bearing+rig	Information about the bearing type and the test rig, ['NU204_E_plastic_a', 'NU204_E_metal_b', 'NU204_E_plastic_b', 'STO20_b']
Data_No	Number of the recording, [1 ... 300]

Data set creator:

Hochschule Esslingen - University of Applied Sciences
Institute for Technical Reliability and Prognostics (IZP)
Robert-Bosch-Straße 1
73037 Göppingen
Germany

Data set citation:

Please refer to the following source when using this data set:

Braig, M.; Zeiler, P. (2025): A Study on Using Transfer Learning to Utilize Information from Similar Systems for Data-Driven Condition Diagnosis and Prognosis. IEEE Access, Volume 13, pp. 98485-98503, DOI: 10.1109/ACCESS.2025.3576435